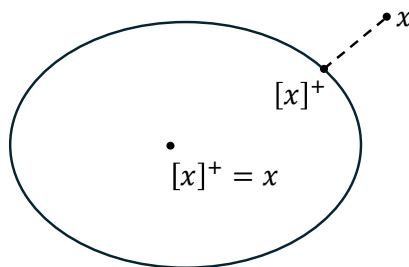


ECE 647: Lecture 26

Projection $[\mathbf{x}]^+$: (for L_2 -norm)

The projection of $\mathbf{x}$ onto the convex and closed set $\mathcal{X}$ is the point $\mathbf{z} \in \mathcal{X}$ that is closest to $\mathbf{x}$, i.e.,

$$[\mathbf{x}]^+ = \underset{\mathbf{z} \in \mathcal{X}}{\operatorname{argmin}} \|\mathbf{z} - \mathbf{x}\|_2$$



Example:

$$1: \mathcal{X} = \bigotimes_{i=1}^m [a_i, b_i]$$

$$[\mathbf{x}]^+ = \{[\mathbf{x}_i]^+\}$$

where

$$[\mathbf{x}_i]^+ = \begin{cases} a_i & \text{if } x_i \leq a_i \\ b_i & \text{if } x_i \geq b_i \\ x_i & \text{otherwise} \end{cases}$$

2: Projection to a Polyhedra (Refer to Boyd, p. 390)

$$\begin{aligned} \min_{\mathbf{x}} \quad & \|\mathbf{x} - \mathbf{x}_0\|_2^2 \\ \text{s.t.} \quad & A\mathbf{x} \leq \mathbf{b} \end{aligned}$$

This is a quadratic program.

Solution:

On a hyperplane $a^T \mathbf{x} = b$

$$P_C(\mathbf{x}_0) = \mathbf{x}_0 + (b - a^T \mathbf{x}_0) \cdot \frac{a}{\|a\|_2^2}$$

On a half-space $a^T \mathbf{x} \leq b$

$$P_C(\mathbf{x}_0) = \begin{cases} \mathbf{x}_0 & \text{if } a^T \mathbf{x}_0 \leq b \\ \mathbf{x}_0 + (b - a^T \mathbf{x}_0) \cdot \frac{a}{\|a\|_2^2} & \text{if } a^T \mathbf{x}_0 > b \end{cases}$$

Question: Can we derive these by optimality conditions?

- 3: In general, $\mathcal{X} = \{\mathbf{x} \mid f_i(\mathbf{x}) \leq 0, h_i(\mathbf{x}) = 0\}$. Then the projection is given by:

$$\begin{aligned} [\mathbf{x}]^+ &= \underset{\mathbf{z}}{\operatorname{argmin}} \quad \|\mathbf{z} - \mathbf{x}\|_2 \\ &\text{s.t.} \quad f_i(\mathbf{z}) \leq 0 \\ &\quad \quad h_i(\mathbf{z}) = 0 \end{aligned}$$

This is not a trivial operation! (Reason to use dual algorithm)

Projection Theorem (Bertsekas & Tsitsiklis, p. 211)

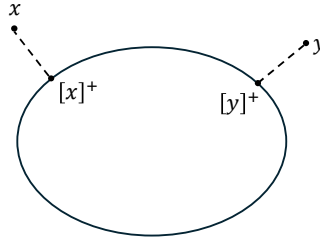
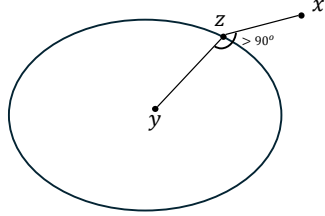
- (a): For every $\mathbf{x} \in \mathbb{R}^n$, there exists a unique $\mathbf{z} \in \mathcal{X}$ that minimizes $\|\mathbf{z} - \mathbf{x}\|_2$ over all $\mathbf{z} \in \mathcal{X}$, and will be denoted as $[\mathbf{x}]^+$.

- (b): Given some $\mathbf{x} \in \mathbb{R}^n$, a vector $\mathbf{z} \in \mathcal{X}$ is equal to $[\mathbf{x}]^+$ if and only if

$$(\mathbf{y} - \mathbf{z})^T (\mathbf{x} - \mathbf{z}) \leq 0 \quad \text{for all } \mathbf{y} \in \mathcal{X}$$

- (c): The mapping $f(\mathbf{x}) = [\mathbf{x}]^+$ is continuous and non-expansive, that is

$$\|[\mathbf{x}]^+ - [\mathbf{y}]^+\|_2 \leq \|\mathbf{x} - \mathbf{y}\|_2 \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$$



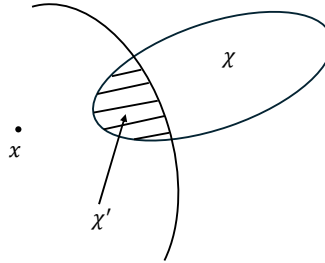
Proof:

(a): Can intersect $\mathcal{X}$ with a closed & bounded set.

Minimize $\|\mathbf{z} - \mathbf{x}\|_2^2$ over $\mathcal{X}'$ (closed, bounded).

$\implies$ exists a minimum point $[\mathbf{x}]^+$.

It is unique because $\|\mathbf{z} - \mathbf{x}\|_2^2$ is strictly convex.



(b): Consider $\min g(\mathbf{z}) = \|\mathbf{z} - \mathbf{x}\|_2^2$ subject to $\mathbf{z} \in \mathcal{X}$.

$g'(\mathbf{z}) = 2(\mathbf{z} - \mathbf{x})$.

By optimality condition, $\mathbf{z}$ is optimal

$$\Leftrightarrow 2(\mathbf{z} - \mathbf{x})^T(\mathbf{y} - \mathbf{z}) \geq 0 \quad \text{for all } \mathbf{y} \in \mathcal{X}$$

$$\Leftrightarrow (\mathbf{y} - \mathbf{z})^T(\mathbf{x} - \mathbf{z}) \leq 0$$

(c): From part (b)

$$([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{x} - [\mathbf{x}]^+) \leq 0$$

Similarly

$$([\mathbf{x}]^+ - [\mathbf{y}]^+)^T(\mathbf{y} - [\mathbf{y}]^+) \leq 0$$

Adding these two inequalities:

$$\begin{aligned} &([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{x} - [\mathbf{x}]^+) + ([\mathbf{x}]^+ - [\mathbf{y}]^+)^T(\mathbf{y} - [\mathbf{y}]^+) \leq 0 \\ &([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{x} - [\mathbf{x}]^+ - (\mathbf{y} - [\mathbf{y}]^+)) \leq 0 \\ &([\mathbf{y}]^+ - [\mathbf{x}]^+)^T((\mathbf{x} - \mathbf{y}) - ([\mathbf{x}]^+ - [\mathbf{y}]^+)) \leq 0 \\ &([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{y} - \mathbf{x}) - ([\mathbf{y}]^+ - [\mathbf{x}]^+)^T([\mathbf{x}]^+ - [\mathbf{y}]^+) \leq 0 \\ &\quad - ([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{x} - \mathbf{y}) - \|[\mathbf{y}]^+ - [\mathbf{x}]^+\|_2^2 \leq 0 \\ &\|[\mathbf{y}]^+ - [\mathbf{x}]^+\|_2^2 \leq -([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{x} - \mathbf{y}) = ([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{y} - \mathbf{x}) \end{aligned}$$

Using Cauchy-Schwarz inequality:

$$\|[\mathbf{y}]^+ - [\mathbf{x}]^+\|_2^2 \leq ([\mathbf{y}]^+ - [\mathbf{x}]^+)^T(\mathbf{y} - \mathbf{x}) \leq \|[\mathbf{y}]^+ - [\mathbf{x}]^+\|_2 \|\mathbf{y} - \mathbf{x}\|_2$$

The result then follows by dividing by $\|[\mathbf{y}]^+ - [\mathbf{x}]^+\|_2$ (if non-zero).

Gradient Projection Algorithm

The gradient projection algorithm is given by the update rule:

$$\mathbf{x}^{(t+1)} = [\mathbf{x}^{(t)} - \gamma \nabla f(\mathbf{x}^{(t)})]^+$$

Lemma: Assume that f is convex and differentiable. $\mathbf{x}^*$ minimizes $f(\mathbf{x})$ over $\mathcal{X}$ if and only if

$$\mathbf{x}^* = [\mathbf{x}^* - \gamma \nabla f(\mathbf{x}^*)]^+$$

Proof: " $\Leftarrow$ " Assume that $\mathbf{x}^* = [\mathbf{x}^* - \gamma \nabla f(\mathbf{x}^*)]^+$. Then, for all $\mathbf{y} \in \mathcal{X}$:

$$\begin{aligned} (\mathbf{y} - \mathbf{x}^*)^T ((\mathbf{x}^* - \gamma \nabla f(\mathbf{x}^*)) - \mathbf{x}^*) &\leq 0 \\ (\mathbf{y} - \mathbf{x}^*)^T (-\gamma \nabla f(\mathbf{x}^*)) &\leq 0 \\ \Leftrightarrow (\mathbf{y} - \mathbf{x}^*)^T \nabla f(\mathbf{x}^*) &\geq 0 \end{aligned}$$

" $\Rightarrow$ " can be shown analogously.

Theorem: If f is convex, ∇f is Lipschitz with parameter L , and there exists at least one point $\mathbf{x}^*$ such that

$$\mathbf{x}^* = [\mathbf{x}^* - \gamma \nabla f(\mathbf{x}^*)]^+,$$

then the sequence of points

$$\mathbf{x}^{(t+1)} = [\mathbf{x}^{(t)} - \gamma \nabla f(\mathbf{x}^{(t)})]^+$$

converges if $0 < \gamma < \frac{2}{L}$, and the limit $\mathbf{x}_0$ minimizes $f(\mathbf{x})$ over $\mathcal{X}$.

Proof: Let $\mathbf{y}^{(t+1)} = \mathbf{x}^{(t)} - \gamma \nabla f(\mathbf{x}^{(t)})$. Let $\mathbf{x}^*$ be one of the optimal points. Let $\mathbf{y}^* = \mathbf{x}^* - \gamma \nabla f(\mathbf{x}^*)$.

Using the techniques in the proof of the standard gradient algorithm, we have

$$\|\mathbf{y}^{(t+1)} - \mathbf{y}^*\|_2^2 \leq \|\mathbf{x}^{(t)} - \mathbf{x}^*\|_2^2 - \left(\frac{2\gamma}{L} - \gamma^2 \right) \|\nabla f(\mathbf{x}^{(t)}) - \nabla f(\mathbf{x}^*)\|_2^2$$

Using the projection theorem, we have

$$\|\mathbf{x}^{(t+1)} - \mathbf{x}^*\|_2^2 \leq \|\mathbf{y}^{(t+1)} - \mathbf{y}^*\|_2^2$$

The rest of the proof follows the case without projection.